Data Analysis and Storytelling

Class introduction

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FIRS - Track 1

Objectives of the course

Learn how to

- perform data analysis to answer questions concerning forest resources
- create visualizations, integrated in a narrative structure, to communicate results from statistical analyses to a broad audience

Overview of the course schedule

- 08/09/2026: data analysis - regression, linear model, extensions, variable and model selection, generalized linear model, hierarchical models
- 09/09/2026: data visualization - graphic display of data, main visualization approaches
- 10/09/2026 am: data analysis - temporal and spatial structures in regression models
- 10/09/2026 pm: data visualization - spatio-temporal data visualization
- 02/11/2026 am: conference by N. Derrière IGN/SIF
- 02/11/2026 pm: uncertainty visualization
- 03/11/2026 am: data storytelling
- 03/11/2026 pm: exam + project launch

Instructor

- Lionel Hertzog - researcher, IGN - Géodata Paris, data analysis
- Jacques Gautier - lecturer, IGN - Géodata Paris, visualization
- Maria Jesus Lobo - researcher, IGN - Géodata Paris, data storytelling
- Nathalie Derrière - engineer, IGN - NFI, NFI results communication

Practicalities

- Class from 08:30 to 11:45 (break between 10:00 and 10:15) and from 13:30 to 16:45 (break between 15:00 and 15:15)
- Classroom Guinier
- Course materials on github (see link sent)
- Course taught in English (except for conference on 02/11)

Evaluation

- 40% of class grade from a MCQ exam on 03/11 pm (1h)
- The exam will only be over materials seen in september - so no questions concerning the uncertainty, the storytelling and the conference
- 60% of class grade from a notebook over a project to be returned by the 15/11/2026
- The project will be individual and given on the 03/11 pm

Feedbacks

- Don't hesitate to ask questions throughout the class
- We will have a feedback session on the 03/11 pm before the project launch
- There will also be a track-wide midway feedback session on the 23/10

Structure of the Data Analysis part

- The data analysis part is divided into 9 thematic units of around 45min
- Each unit will consist of (i) slides introducing the topic (10-15min), (ii) a live coding session (10-15min), (iii) a practical, hands-on, session (10-25min)

Structure of the Data Visualization part

- The data visualization part is divided into 6 sessions of three hours
- 5 sessions will each consist of (i) one lecture part (around 1 hour), and (ii) one practical part (2 hours)
- 1 session will consist of a 3-hour conference with a domain expert.